

Dataset Description

Dataset on droplet spreading and rebound behavior of water and viscous water-glycerol mixtures on superhydrophobic surfaces with laser-made channels

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This dataset is organized within 5 files and 1 folder.

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File 1: Rebound Data

Description: This file contains a table consisting of 1498 water-glycerol droplet impact experiments on monolayer-functionalized laser-textured aluminum samples, where each row represents a single measurement and includes the following parameters:

- a) sample name,
- b) droplet diameter (in m),
- c) droplet impact velocity (in m s^{-1}),
- d) density of the droplet (in kg m^{-3}),
- e) surface tension of the droplet (in N m^{-1}),
- f) dynamic viscosity of the droplet (in $\text{Pa}\cdot\text{s}$),
- g) distance between adjacent laser beam passes (in μm),
- h) approximate average depth of laser-made channels (in μm),
- i) Reynolds number (dimensionless),
- j) Weber number (dimensionless),
- k) droplet-surface contact time (in seconds),
- l) rebound efficiency (dimensionless),
- m) droplet maximum spreading factor (dimensionless), and
- n) maximum lamella velocity during the spreading phase (in m s^{-1}).

Type of data: Processed data.

Data format: Data in .csv.

How data were acquired: The superhydrophobic surfaces were prepared on aluminum alloy (1050A H24, $\geq 99.5\%$ Al) plates via two-step fabrication process comprising laser texturing and hydrophobization. A nanosecond fiber laser system (FL Mark-C with JPT Opto-electronics "M7 30 W" MOPA source) was used to fabricate 6 and 25 μm deep microchannels. The laser beam was scanned in a crosshatch pattern with a center-to-center distance of 50, 100, 200, 400, 600 and 800 μm distance between successive passes. Following laser texturing, the surfaces were hydrophobized by drop-casting a 3 mM solution of 1H,1H',2H,2H'-perfluorododecyl-1-phosphonic acid (FDPA) in isopropanol.

Droplet impact measurements were conducted using water-glycerol binary mixtures with 0, 20, 60, 78 and 91 wt.% glycerol. The mixtures were infused through a blunt G27 needle with a syringe pump (Harvard Apparatus PHD 2000) operated at a constant flow rate of $150 \mu\text{L min}^{-1}$ to produce droplet of controlled size. The droplets were released onto the samples from five different heights to vary the impact velocities. Droplet impact was captured using a high-speed optical camera (Photron FASTCAM Mini UX100) equipped with a macro lens (Laowa 100mm f/2.8 2x Ultra Macro APO) at a frame rate of 5000 fps and spatial resolution of $\sim 12 \mu\text{m px}^{-1}$. A white LED backlight (Thorlabs SOLIS-2C

powered by a DC2200 controller) was used for illumination. Prior to each measurement, the temperature was recorded using a SparkFun SHTC3 (Qwiic) sensor and a SparkFun RedBoard Qwiic Arduino controller.

Data source location:

Faculty of Mechanical Engineering, University of Ljubljana, Aškerčeva cesta 6, 1000 Ljubljana, Slovenia.

The data were acquired in the period between Apr – Dec 2024.

File 2: Rebound Data - REF-H

Description:	This file contains a table consisting of 125 water-glycerol droplet impact experiments on a monolayer-functionalized smooth aluminum sample, where each row represents a single measurement and includes the following parameters: a) sample name, b) droplet diameter (in m), c) droplet impact velocity (in m s^{-1}), d) density of the droplet (in kg m^{-3}), e) surface tension of the droplet (in N m^{-1}), f) dynamic viscosity of the droplet (in $\text{Pa}\cdot\text{s}$), g) Reynolds number (dimensionless), h) Weber number (dimensionless), i) droplet maximum spreading factor (dimensionless), and j) maximum lamella velocity during the spreading phase (in m s^{-1}).
Type of data:	Processed data.
Data format:	Data in .csv.
How data were acquired:	The samples were prepared using aluminum alloy (1050A H24, $\geq 99.5\%$ Al) plates, which were hydrophobized by drop-casting a 3 mM solution of 1H,1H',2H,2H'-perfluorododecil-1-phosphonic acid (FDPA) in isopropanol. Droplet impact measurements were conducted using water-glycerol binary mixtures with 0, 20, 60, 78 and 91 wt.% glycerol. The mixtures were infused through a blunt G27 needle with a syringe pump (Harvard Apparatus PHD 2000) operated at a constant flow rate of $150 \mu\text{L min}^{-1}$ to produce droplet of controlled size. The droplets were released onto the samples from five different heights to vary the impact velocities. Droplet impact was captured using a high-speed optical camera (Photron FASTCAM Mini UX100) equipped with a macro lens (Laowa 100mm f/2.8 2x Ultra Macro APO) at a frame rate of 5000 fps and spatial resolution of $\sim 12 \mu\text{m px}^{-1}$. A white LED backlight (Thorlabs SOLIS-2C powered by a DC2200 controller) was used for illumination. Prior to each measurement, the temperature was recorded using a SparkFun SHTC3 (Qwiic) sensor and a SparkFun RedBoard Qwiic Arduino controller.
Data source location:	Faculty of Mechanical Engineering, University of Ljubljana, Aškerčeva cesta 6, 1000 Ljubljana, Slovenia. The data were acquired in the period between Apr – Dec 2024.

File 3: Contact angles - Water

Description:	The file contains a table presenting the average measured static and dynamic contact angles of water droplets on monolayer-functionalized laser-textured aluminum samples featuring 6 and 25 μm deep microchannels. The center-to-center distances between adjacent microchannels are approximately 50, 100, 200, 400, 600 and 800 μm. The table's columns list sample names and the rows list the corresponding static, advancing, and receding contact angles, and contact angle hysteresis.
Type of data:	Processed data.
Data format:	Data in .csv.
How data were acquired:	Static contact angles were measured using a goniometer (Ossila) by depositing water droplets onto the surfaces and observing the resulting contact angle. Each measurement was repeated 5 times. Processing of the videos was performed in Ossila's proprietary software. Advancing and receding contact angles were measured using the droplet inflation/deflation method on a custom-made experimental setup for goniometry equipped with a CMOS camera and a strong LED backlight. A syringe pump was used to inflate and deflate the droplets while the movement of the triple contact line across the surface was recorded with the camera. Processing of the videos was performed using custom-developed algorithms in MathWorks MATLAB environment.
Data source location:	Faculty of Mechanical Engineering, University of Ljubljana, Aškerčeva cesta 6, 1000 Ljubljana, Slovenia. The data were acquired in the period between Apr – Dec 2024.

File 4: Contact angles - Water-Glycerol_91wtPerc

Description:	The file contains a table presenting the average measured static contact angles of water-glycerol droplets with 91 wt.% glycerol on monolayer-functionalized laser-textured aluminum samples featuring 6 and 25 μm deep microchannels. The center-to-center distances between adjacent microchannels are approximately 50, 100, 200, 400, 600 and 800 μm. The table's columns list sample names and a single row lists the corresponding static contact angles.
Type of data:	Processed data.
Data format:	Data in .csv.
How data were acquired:	Static contact angles were measured using a goniometer (Ossila) by depositing droplets of water-glycerol binary mixture with 91 wt.% glycerol onto the surfaces and observing the resulting contact angle. Each measurement was repeated 5 times. Processing of the videos was performed in Ossila's proprietary software.
Data source location:	Faculty of Mechanical Engineering, University of Ljubljana, Aškerčeva cesta 6, 1000 Ljubljana, Slovenia. The data were acquired in the period between Apr – Dec 2024.

File 5: Advancing contact angles - Water-glycerol mixtures

Description: The file contains a table presenting the average measured advancing contact angles of water-glycerol droplets with 20, 60, 78 and 91 wt.% glycerol on monolayer-functionalized laser-textured aluminum samples featuring microchannels with depths of 6 and 25 μm . The center-to-center distances between adjacent microchannels are approximately 50, 200 and 800 μm .

The table's rows list sample names and the corresponding advancing contact angles for the water-glycerol mixtures in the following order: 20, 60, 78 and 91 wt.% glycerol.

Type of data: Processed data.

Data format: Data in .csv.

How data were acquired: Advancing contact angles were measured using the droplet inflation method on a custom-made experimental setup for goniometry equipped with a CMOS camera and a strong LED backlight. A syringe pump was used to inflate the droplets while the movement of the triple contact line across the surface was recorded with the camera. Advancing contact angles were measured using water-glycerol binary mixtures with 20, 60, 78 and 91 wt.% glycerol. Processing of the videos was performed using custom-developed algorithms in MathWorks MATLAB environment.

Data source location: Faculty of Mechanical Engineering, University of Ljubljana, Aškerčeva cesta 6, 1000 Ljubljana, Slovenia.

The data were acquired in the period between Apr – Dec 2024.

Folder 1: SEM images of the test surfaces

Description:	The folder contains 39 scanning electron microscopy (SEM) images of the monolayer-functionalized laser-textured aluminum samples featuring microchannels with depths of 6 and 25 μm and center-to-center distances of approximately 50, 100, 200, 400, 600 and 800 μm, and the monolayer-functionalized smooth aluminum sample. Each sample is imaged at 43$\times$, 100$\times$ and 350$\times$ magnification. The name of each image is composed of the sample name and the magnification of the image.
Type of data:	Raw data.
Data format:	Data in <i>.tif</i> .
How data were acquired:	SEM imaging was performed using a JCM-7000 NeoScope SEM, utilizing the secondary electron detector and operating at an accelerating voltage of 15 kV.
Data source location:	Faculty of Mechanical Engineering, University of Ljubljana, Aškerčeva cesta 6, 1000 Ljubljana, Slovenia. The data were acquired in the period between Apr – Dec 2024.